# 配置集群间 LIF

## ONTAP 9

NetApp
January 28, 2022

This PDF was generated from https://docs.netapp.com/zh-cn/ontap/peering/configure-intercluster-lifs-share-data-ports-task.html on January 28, 2022. Always check docs.netapp.com for the latest.

# 目录

配置集群间 LIF . . . . . 1
在共享数据端口上配置集群间 LIF . . . . . 1
在专用端口上配置集群间 LIF . . . . . 4
在自定义 IP 空间中配置集群间 LIF . . . . . 8

# 配置集群间 LIF

## 在共享数据端口上配置集群间 LIF

您可以在与数据网络共享的端口上配置集群间 LIF 。这样可以减少集群间网络连接所需的端口数量。

步骤

1. 列出集群中的端口：

   ```
   network port show
   ```

   有关完整的命令语法，请参见手册页。

   以下示例显示了 `cluster01` 中的网络端口：

   ```
   cluster01::> network port show
                                                                     Speed
   (Mbps)
   Node   Port      IPspace      Broadcast Domain Link   MTU    Admin/Oper
   ------ --------- ------------ ---------------- ----- -------
   ------------
   cluster01-01
          e0a       Cluster      Cluster          up     1500   auto/1000
          e0b       Cluster      Cluster          up     1500   auto/1000
          e0c       Default      Default          up     1500   auto/1000
          e0d       Default      Default          up     1500   auto/1000
   cluster01-02
          e0a       Cluster      Cluster          up     1500   auto/1000
          e0b       Cluster      Cluster          up     1500   auto/1000
          e0c       Default      Default          up     1500   auto/1000
          e0d       Default      Default          up     1500   auto/1000
   ```

2. 在系统 SVM 上创建集群间 LIF ：

| 选项 | Description |
| --- | --- |
| • 在 ONTAP 9.6 及更高版本中： * | `network interface create -vserver` *`system_svm`* `-lif` *`LIF_name`* `-service -policy default-intercluster -home -node` *`node`*`-home-port port -address` *`port_ip`* `-netmask` *`netmask`* |

| 选项 | Description |
| --- | --- |
| • 在 ONTAP 9.5 及更早版本中： * | `network interface create -vserver` *`system_svm`* `-lif` *`LIF_name`* `-role intercluster -home-node` *`node`*`-home-port` *`port`* `-address` *`port_ip`* `-netmask` *`netmask`* |

有关完整的命令语法，请参见手册页。

以下示例将创建集群间 LIF `cluster01_icl01` 和 `cluster01_icl02`：

```
cluster01::> network interface create -vserver cluster01 -lif
cluster01_icl01 -service-
policy default-intercluster -home-node cluster01-01 -home-port e0c
-address 192.168.1.201
-netmask 255.255.255.0

cluster01::> network interface create -vserver cluster01 -lif
cluster01_icl02 -service-
policy default-intercluster -home-node cluster01-02 -home-port e0c
-address 192.168.1.202
-netmask 255.255.255.0
```

3. 验证是否已创建集群间 LIF：

| 选项 | Description |
| --- | --- |
| • 在 ONTAP 9.6 及更高版本中： * | `network interface show -service-policy default-intercluster` |
| • 在 ONTAP 9.5 及更早版本中： * | `network interface show -role intercluster` |

有关完整的命令语法，请参见手册页。

```
cluster01::> network interface show -service-policy default-intercluster
            Logical    Status     Network            Current
Current Is
Vserver     Interface  Admin/Oper Address/Mask       Node          Port
Home
----------- ---------- ---------- ------------------ -------------
------- ----
cluster01
            cluster01_icl01
                       up/up      192.168.1.201/24   cluster01-01  e0c
true
            cluster01_icl02
                       up/up      192.168.1.202/24   cluster01-02  e0c
true
```

4. 验证集群间 LIF 是否冗余：

| 选项 | Description |
| --- | --- |
| • 在 ONTAP 9.6 及更高版本中： * | `network interface show – service-policy default-intercluster -failover` |
| • 在 ONTAP 9.5 及更早版本中： * | `network interface show -role intercluster -failover` |

有关完整的命令语法，请参见手册页。

以下示例显示 `e0c` 端口上的集群间 LIF `cluster01_icl01` 和 `cluster01_icl02` 将故障转移到 `e0d` 端口。

```
cluster01::> network interface show -service-policy default-intercluster
-failover
         Logical         Home                  Failover        Failover
Vserver  Interface       Node:Port             Policy          Group
-------- --------------- --------------------- --------------- --------
cluster01
         cluster01_icl01 cluster01-01:e0c   local-only
192.168.1.201/24
                            Failover Targets: cluster01-01:e0c,
                                              cluster01-01:e0d
         cluster01_icl02 cluster01-02:e0c   local-only
192.168.1.201/24
                            Failover Targets: cluster01-02:e0c,
                                              cluster01-02:e0d
```

# 在专用端口上配置集群间 LIF

您可以在专用端口上配置集群间 LIF 。这样做通常会增加复制流量的可用带宽。

步骤

1. 列出集群中的端口：

   `network port show`

   有关完整的命令语法，请参见手册页。

   以下示例显示了 `cluster01` 中的网络端口：

   ```
   cluster01::> network port show
                                                                  Speed
   (Mbps)
   Node   Port      IPspace      Broadcast Domain Link   MTU    Admin/Oper
   ------ --------- ------------ ---------------- ----- -------
   ------------
   cluster01-01
          e0a       Cluster      Cluster          up     1500   auto/1000
          e0b       Cluster      Cluster          up     1500   auto/1000
          e0c       Default      Default          up     1500   auto/1000
          e0d       Default      Default          up     1500   auto/1000
          e0e       Default      Default          up     1500   auto/1000
          e0f       Default      Default          up     1500   auto/1000
   cluster01-02
          e0a       Cluster      Cluster          up     1500   auto/1000
          e0b       Cluster      Cluster          up     1500   auto/1000
          e0c       Default      Default          up     1500   auto/1000
          e0d       Default      Default          up     1500   auto/1000
          e0e       Default      Default          up     1500   auto/1000
          e0f       Default      Default          up     1500   auto/1000
   ```

2. 确定哪些端口可专用于集群间通信：

   `network interface show -fields home-port ， curr-port`

   有关完整的命令语法，请参见手册页。

   以下示例显示未为端口 `e0e` 和 `e0f` 分配 LIF ：

```
cluster01::> network interface show -fields home-port,curr-port
vserver lif                  home-port curr-port
------- -------------------- --------- ---------
Cluster cluster01-01_clus1   e0a       e0a
Cluster cluster01-01_clus2   e0b       e0b
Cluster cluster01-02_clus1   e0a       e0a
Cluster cluster01-02_clus2   e0b       e0b
cluster01
        cluster_mgmt         e0c       e0c
cluster01
        cluster01-01_mgmt1   e0c       e0c
cluster01
        cluster01-02_mgmt1   e0c       e0c
```

3. 为专用端口创建故障转移组：

   `network interface failover-groups create -vserver` *`system_svm`* `-failover-group` *`failover_group`* `-targets` *`physical_or_logical_ports`*

   以下示例将端口 `e0e` 和 `e0f` 分配给系统 SVM `cluster01` 上的故障转移组 `intercluster01`：

```
cluster01::> network interface failover-groups create -vserver cluster01
-failover-group
intercluster01 -targets
cluster01-01:e0e,cluster01-01:e0f,cluster01-02:e0e,cluster01-02:e0f
```

4. 验证是否已创建故障转移组：

   `network interface failover-groups show`

   有关完整的命令语法，请参见手册页。

```
cluster01::> network interface failover-groups show
                                  Failover
Vserver          Group            Targets
---------------- ----------------
--------------------------------------------
Cluster
                 Cluster
                                  cluster01-01:e0a, cluster01-01:e0b,
                                  cluster01-02:e0a, cluster01-02:e0b
cluster01
                 Default
                                  cluster01-01:e0c, cluster01-01:e0d,
                                  cluster01-02:e0c, cluster01-02:e0d,
                                  cluster01-01:e0e, cluster01-01:e0f
                                  cluster01-02:e0e, cluster01-02:e0f
                 intercluster01
                                  cluster01-01:e0e, cluster01-01:e0f
                                  cluster01-02:e0e, cluster01-02:e0f
```

5. 在系统 SVM 上创建集群间 LIF 并将其分配给故障转移组。

| 选项 | Description |
| --- | --- |
| • 在 ONTAP 9.6 及更高版本中： * | `network interface create -vserver` *`system_svm`* `-lif` *`LIF_name`* `-service -policy default-intercluster -home -node` *`node`*`-home-port` *`port`* `-address` *`port_ip`* `-netmask` *`netmask`* `-failover -group` *`failover_group`* |
| • 在 ONTAP 9.5 及更早版本中： * | `network interface create -vserver` *`system_svm`* `-lif` *`LIF_name`* `-role intercluster -home-node` *`node`*`-home-port` *`port`* `-address` *`port_ip`* `-netmask` *`netmask`* `-failover-group` *`failover_group`* |

有关完整的命令语法，请参见手册页。

以下示例将在故障转移组 `intercluster01` 中创建集群间 LIF `cluster01_icl01` 和 `cluster01_icl02`：

```
cluster01::> network interface create -vserver cluster01 -lif
cluster01_icl01 -service-
policy default-intercluster -home-node cluster01-01 -home-port e0e
-address 192.168.1.201
-netmask 255.255.255.0 -failover-group intercluster01

cluster01::> network interface create -vserver cluster01 -lif
cluster01_icl02 -service-
policy default-intercluster -home-node cluster01-02 -home-port e0e
-address 192.168.1.202
-netmask 255.255.255.0 -failover-group intercluster01
```

6. 验证是否已创建集群间 LIF：

| 选项 | Description |
|---|---|
| • 在 ONTAP 9.6 及更高版本中： * | `network interface show -service-policy default-intercluster` |
| • 在 ONTAP 9.5 及更早版本中： * | `network interface show -role intercluster` |

有关完整的命令语法，请参见手册页。

```
cluster01::> network interface show -service-policy default-intercluster
            Logical    Status     Network            Current
Current Is
Vserver     Interface  Admin/Oper Address/Mask       Node           Port
Home
----------- ---------- ---------- ------------------ -------------
------- ----
cluster01
            cluster01_icl01
                       up/up      192.168.1.201/24   cluster01-01  e0e
true
            cluster01_icl02
                       up/up      192.168.1.202/24   cluster01-02  e0f
true
```

7. 验证集群间 LIF 是否冗余：

| 选项 | Description |
| --- | --- |
| • 在 ONTAP 9.6 及更高版本中： * | `network interface show -service-policy default-intercluster -failover` |
| • 在 ONTAP 9.5 及更早版本中： * | `network interface show -role intercluster -failover` |

有关完整的命令语法，请参见手册页。

以下示例显示 SVM`e0e` 端口上的集群间 LIF `cluster01_icl01` 和 `cluster01_icl02` 将故障转移到 `e0f` 端口。

```
cluster01::> network interface show -service-policy default-intercluster
-failover
         Logical         Home                  Failover        Failover
Vserver  Interface       Node:Port             Policy          Group
-------- --------------- --------------------- --------------- --------
cluster01
         cluster01_icl01 cluster01-01:e0e   local-only
intercluster01
                            Failover Targets:  cluster01-01:e0e,
                                               cluster01-01:e0f
         cluster01_icl02 cluster01-02:e0e   local-only
intercluster01
                            Failover Targets:  cluster01-02:e0e,
                                               cluster01-02:e0f
```

# 在自定义 IP 空间中配置集群间 LIF

## 您可以在自定义 IP 空间中配置集群间 LIF 。这样可以在多租户环境中隔离复制流量。

创建自定义 IP 空间时，系统会创建一个系统 Storage Virtual Machine （ SVM ），用作该 IP 空间中系统对象的容器。您可以使用新的 SVM 作为新 IP 空间中任何集群间 LIF 的容器。新的 SVM 与自定义 IP 空间同名。

步骤

1. 列出集群中的端口：

   `network port show`

   有关完整的命令语法，请参见手册页。

   以下示例显示了 `cluster01` 中的网络端口：

```
cluster01::> network port show
                                                                  Speed
(Mbps)
Node   Port      IPspace      Broadcast Domain Link   MTU    Admin/Oper
------ --------- ------------ ---------------- ----- -------
------------
cluster01-01
       e0a       Cluster      Cluster          up     1500   auto/1000
       e0b       Cluster      Cluster          up     1500   auto/1000
       e0c       Default      Default          up     1500   auto/1000
       e0d       Default      Default          up     1500   auto/1000
       e0e       Default      Default          up     1500   auto/1000
       e0f       Default      Default          up     1500   auto/1000
cluster01-02
       e0a       Cluster      Cluster          up     1500   auto/1000
       e0b       Cluster      Cluster          up     1500   auto/1000
       e0c       Default      Default          up     1500   auto/1000
       e0d       Default      Default          up     1500   auto/1000
       e0e       Default      Default          up     1500   auto/1000
       e0f       Default      Default          up     1500   auto/1000
```

2. 在集群上创建自定义 IP 空间：

   `network ipspace create -ipspace` *`ipspace`*

   以下示例将创建自定义 IP 空间 `ipspace-IC1`：

   ```
   cluster01::> network ipspace create -ipspace ipspace-IC1
   ```

3. 确定哪些端口可专用于集群间通信：

   `network interface show -fields home-port ， curr-port`

   有关完整的命令语法，请参见手册页。

   以下示例显示未为端口 `e0e` 和 `e0f` 分配 LIF：

```
cluster01::> network interface show -fields home-port,curr-port
vserver lif                  home-port curr-port
------- -------------------- --------- ---------
Cluster cluster01_clus1   e0a       e0a
Cluster cluster01_clus2   e0b       e0b
Cluster cluster02_clus1   e0a       e0a
Cluster cluster02_clus2   e0b       e0b
cluster01
        cluster_mgmt         e0c       e0c
cluster01
        cluster01-01_mgmt1   e0c       e0c
cluster01
        cluster01-02_mgmt1   e0c       e0c
```

4. 从默认广播域中删除可用端口：

   `network port broadcast-domain remove-ports -broadcast-domain default -ports ports ports`

   一个端口一次不能位于多个广播域中。有关完整的命令语法，请参见手册页。

   以下示例将从默认广播域中删除端口 `e0e` 和 `e0f`：

```
cluster01::> network port broadcast-domain remove-ports -broadcast
-domain Default -ports
cluster01-01:e0e,cluster01-01:e0f,cluster01-02:e0e,cluster01-02:e0f
```

5. 验证端口是否已从默认广播域中删除：

   `network port show`

   有关完整的命令语法，请参见手册页。

   以下示例显示已从默认广播域中删除端口 `e0e` 和 `e0f`：

```
cluster01::> network port show
                                                          Speed (Mbps)
Node   Port    IPspace   Broadcast Domain Link   MTU    Admin/Oper
------ ------- --------- --------------- ----- ------- ------------
cluster01-01
       e0a     Cluster    Cluster          up    9000  auto/1000
       e0b     Cluster    Cluster          up    9000  auto/1000
       e0c     Default    Default          up    1500  auto/1000
       e0d     Default    Default          up    1500  auto/1000
       e0e     Default    -                up    1500  auto/1000
       e0f     Default    -                up    1500  auto/1000
       e0g     Default    Default          up    1500  auto/1000
cluster01-02
       e0a     Cluster    Cluster          up    9000  auto/1000
       e0b     Cluster    Cluster          up    9000  auto/1000
       e0c     Default    Default          up    1500  auto/1000
       e0d     Default    Default          up    1500  auto/1000
       e0e     Default    -                up    1500  auto/1000
       e0f     Default    -                up    1500  auto/1000
       e0g     Default    Default          up    1500  auto/1000
```

6. 在自定义 IP 空间中创建广播域：

   `network port broadcast-domain create -ipspace` *`ipspace`*`-broadcast-domain` *`broadcast_domain`* `-mtu` *`mtu _ -ports _ports`*

   以下示例将在 IP 空间 `ipspace-IC1` 中创建广播域 `ipspace-IC1-bd`：

```
cluster01::> network port broadcast-domain create -ipspace ipspace-IC1
-broadcast-domain
ipspace-IC1-bd -mtu 1500 -ports cluster01-01:e0e,cluster01-01:e0f,
cluster01-02:e0e,cluster01-02:e0f
```

7. 验证是否已创建广播域：

   `network port broadcast-domain show`

   有关完整的命令语法，请参见手册页。

```
cluster01::> network port broadcast-domain show
IPspace Broadcast                                         Update
Name    Domain Name    MTU  Port List                     Status Details
------- ----------- ------  ------------------------------ --------------
Cluster Cluster       9000
                            cluster01-01:e0a              complete
                            cluster01-01:e0b              complete
                            cluster01-02:e0a              complete
                            cluster01-02:e0b              complete
Default Default       1500
                            cluster01-01:e0c              complete
                            cluster01-01:e0d              complete
                            cluster01-01:e0f              complete
                            cluster01-01:e0g              complete
                            cluster01-02:e0c              complete
                            cluster01-02:e0d              complete
                            cluster01-02:e0f              complete
                            cluster01-02:e0g              complete
ipspace-IC1
        ipspace-IC1-bd
                      1500
                            cluster01-01:e0e              complete
                            cluster01-01:e0f              complete
                            cluster01-02:e0e              complete
                            cluster01-02:e0f              complete
```

8. 在系统 SVM 上创建集群间 LIF 并将其分配给广播域：

| 选项 | Description |
|---|---|
| • 在 ONTAP 9.6 及更高版本中： * | `network interface create -vserver` *`system_svm`* `-lif` *`LIF_name`* `-service -policy default-intercluster -home -node` *`node`*`-home-port` *`port`* `-address` *`port_ip`* `-netmask` *`netmask`* |
| • 在 ONTAP 9.5 及更早版本中： * | `network interface create -vserver` *`system_svm`* `-lif` *`LIF_name`* `-role intercluster -home-node` *`node`*`-home-port` *`port`* `-address` *`port_ip`* `-netmask` *`netmask`* |

此时将在分配主端口的广播域中创建 LIF 。此广播域具有一个默认故障转移组，其名称与此广播域相同。有关完整的命令语法，请参见手册页。

以下示例将在广播域 `ipspace-IC1-bd` 中创建集群间 LIF `cluster01_icl01` 和 `cluster01_icl02`：

```
cluster01::> network interface create -vserver ipspace-IC1 -lif
cluster01_icl01 -service-
policy default-intercluster -home-node cluster01-01 -home-port e0e
-address 192.168.1.201
-netmask 255.255.255.0

cluster01::> network interface create -vserver ipspace-IC1 -lif
cluster01_icl02 -service-
policy default-intercluster -home-node cluster01-02 -home-port e0e
-address 192.168.1.202
-netmask 255.255.255.0
```

9. 验证是否已创建集群间 LIF：

| 选项 | Description |
|---|---|
| • 在 ONTAP 9.6 及更高版本中： * | `network interface show -service-policy default-intercluster` |
| • 在 ONTAP 9.5 及更早版本中： * | `network interface show -role intercluster` |

有关完整的命令语法，请参见手册页。

```
cluster01::> network interface show -service-policy default-intercluster
            Logical    Status     Network            Current
Current Is
Vserver     Interface  Admin/Oper Address/Mask       Node           Port
Home
----------- ---------- ---------- ------------------ -------------
------- ----
ipspace-IC1
            cluster01_icl01
                       up/up      192.168.1.201/24   cluster01-01  e0e
true
            cluster01_icl02
                       up/up      192.168.1.202/24   cluster01-02  e0f
true
```

10. 验证集群间 LIF 是否冗余：

| 选项 | Description |
| --- | --- |
| • 在 ONTAP 9.6 及更高版本中： * | `network interface show -service-policy default-intercluster -failover` |
| • 在 ONTAP 9.5 及更早版本中： * | `network interface show -role intercluster -failover` |

有关完整的命令语法，请参见手册页。

以下示例显示 SVM `e0e` 端口上的集群间 LIF `cluster01_icl01` 和 `cluster01_icl02` 故障转移到`e0f`端口：

```
cluster01::> network interface show -service-policy default-intercluster
-failover
         Logical         Home                  Failover        Failover
Vserver  Interface       Node:Port             Policy          Group
-------- --------------- --------------------- --------------- --------
ipspace-IC1
         cluster01_icl01 cluster01-01:e0e   local-only
intercluster01
                            Failover Targets:  cluster01-01:e0e,
                                               cluster01-01:e0f
         cluster01_icl02 cluster01-02:e0e   local-only
intercluster01
                            Failover Targets:  cluster01-02:e0e,
                                               cluster01-02:e0f
```

**Copyright Information**

Copyright © 2022 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system- without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

**Trademark Information**

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.